

Mehrdad Momeni Zadeh

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TECHNICAL SKILLS

Languages: Java, Python, C++, Go, TypeScript, SQL, Kotlin, R
Web and Backend: Spring Boot, Hibernate, Next.js, React, Django, RESTful APIs
DevOps and Cloud: AWS (S3), GCP, Docker, Kubernetes, Jenkins, Git, CI/CD, JWT
ML and Data: PyTorch, scikit-learn, OpenCV, WandB, Hydra, NumPy, Pandas
Databases: PostgreSQL, MySQL, MongoDB, Firebase
Testing Frameworks: JUnit 5, Mockito, pytest, gomock

WORK EXPERIENCE

DeLTA Lab — Simon Fraser University

Burnaby, BC

Undergraduate Researcher — Applied Machine Learning (🔗 [Code](#))

Jan 2025 – Aug 2025

- Built and deployed **ML training pipelines** using **Docker** on **Google Cloud**, reducing setup time by **30%**.
- Helped establish automated **CI/CD workflows** in a **6-person** team, reducing integration bugs by **20%**.
- Optimized model runs to achieve **4× faster training** and **12% higher accuracy** on large-scale datasets.
- Worked with researchers and engineers to develop scalable **data-processing tools**.
- Contributed to a paper accepted to **ICLR 2026**, strengthening technical writing and teamwork skills.

TECHNICAL PROJECTS

Stage Pass — Full-stack event ticketing system for university events.

[GitHub](#) 📄

Tech: Java, Spring Boot 3, TypeScript, PostgreSQL, React, REST APIs

- Deployed for **SFU events**, processing student tickets with zero errors.
- Implemented secure organizer and attendee flows using **Spring Security** with role-based access.
- Collaborated in a **team of three** to design **REST APIs** and database schema for event management.

Micro Gopher — Distributed microservices platform in Go.

[GitHub](#) 📄

Tech: Go, Docker, Kubernetes, PostgreSQL, MongoDB, gRPC

- Refactored a monolithic app into modular **Go microservices** to improve scalability.
- Developed microservices for authentication, logging, and mailing communicating via **gRPC**.
- Containerized and deployed the system with **Docker** and **Kubernetes** for testing in distributed environments.

Galactic Blitz — 2D space shooter built on a custom C++ engine.

[GitHub](#) 📄

Tech: C++, SFML, Box2D, CMake

- Built a simple **C++ game engine** supporting real-time rendering and physics simulation.
- Achieved smooth **60 FPS gameplay** by optimizing rendering and collision detection.
- Structured the engine code into clear modules, making it easier to add new features and fix bugs.

Neuro Driver — 2D driving simulator with neural agents.

[GitHub](#) 📄

Tech: Python, Box2D, Genetic Algorithms, NumPy

- Trained neural agents to reduce collisions by **20%** over **50 generations**.
- Increased lap completion rates by **30%** through reward-tuned training.
- Automated training runs to make experiments easy to track and repeat.

EDUCATION

B.Sc. in Computer Science — GPA: 3.4/4.33

Expected Sep 2027

Simon Fraser University

Burnaby, BC

- **Relevant Coursework:** Data Structures and Algorithms, Computer Systems, Databases, Data Communications and Networking, Machine Learning, Deep Learning, Affective Computing